

National Exam of Higher National Diploma 2022 Session

Spécialty/option: SWE, NWS, EDM

Paper: Discrete Mathematics

Duration : 4 Hours

Credits : 4/4/10

GENERAL INSTRUCTIONS.

*You are reminded of the necessity of good English and orderly presentation of your material.
Where calculations are required, clearly show your working and be chronological in your answer.*

SECTION A :MCQS (20 MARKS)

INSTRUCTIONS:

Each question has four suggested answers: A, B, C, and D. Select the letter of the correct answer.

1) The range of the data 5, 15, 20, 3, 1, -5, and 17 is

- A) 25 B) 19 C) 15 D) 16

2) Given that the mean of 4 numbers is 12 and when the fifth number is added, the mean becomes 15, the fifth number is:

- A) 60 B) 30 C) 15 D) 27

3) The following information is known about a sample: $n = 100$, $\sum x = 200$, $\sum x^2 = 600$.
The
variance of the sample is :

- A) 2 B) 4 C) 5 D) 6

4) The lower class limit of the class 10 - 19 in the data 0 - 9, 10 - 19, 20 - 29 is:

- A) 10.5 B) 10 C) 9 D) 9.5

5) The stem and leaf plot of a data set is shown below.

Stem	leaves
5	5 4 1
6	6 4 3 5 4
7	8 4 1 4 5 8 9 1
8	2 8 0 2 4 3
9	4 1 6

This diagram reveals that the population from which the data is drawn is

- A) Normal B) Skewed to the right C) Skewed to the left D) None of the above

6) The probability density function, f , of a random variable X is defined by

$$f(x) = \begin{cases} 6x(1-x), & 0 < x < 1 \\ 0, & \text{otherwise.} \end{cases}$$

$$P(X < .05) =$$

- A) 0.25 B) 0.5 C) 0.75 D) 0.4

7) If $P(A) = 0.75$, $P(B) = 0.4$, and $P(A \text{ or } B) = 0.9$, $P(A \text{ and } B)$ is

- A) 0.75 B) 0.4 C) 0.65 D) 0.25

8) When a computer goes down there is a 75% chance that it is due to an overload and a 15% chance that it is due to a software problem. There is an 85% chance that it is due to an overload or a software problem. the probability that the breakdown is only due to a software problem is

- A) 0.15 B) 0.7 C) 0.1 D) 0.15

9) A batch of 25 injection-molded parts contains 5 that have suffered excessive shrinkage. Two parts are selected at random without replacement, The probability that the second part selected is one with excessive shrinkage is

- A) $\frac{1}{5}$ B) $\frac{1}{6}$ C) $\frac{1}{30}$ D) $\frac{4}{5}$

- 10) A computer center has three printers A, B and C, which print at different speeds. Programs are routed to the first available printer. The probability that a program is routed to printers A, B and C are 0.6, 0.3 and 0.1, respectively. Occasionally, a printer will jam and destroy a printout. The probability that printers A, B and C will jam are 0.01, 0.05, and 0.04, respectively. Your program is destroyed when a printer jams. What is the probability that printer A is involved?
- A) 0.006 B) 0.24 C) 0.025 D) 0.01
- 11) Given that $(x+2)$ is a factor of the polynomial $f(x) = x^3 - 2x^2 + kx + 2$. The value of K is
- A) 6 B) 1 C) 7 D) -7
- 12) The coefficient of the term in x^2 in the Maclaurin expansion of $\ln(1 - 2x)$ is
- A) 4 B) 1 C) 2 D) -2
- 13) $\lim_{x \rightarrow \infty} \frac{\sqrt{2x^2+1}}{3x-5} =$
- A) $\frac{2}{3}$ B) $\frac{\sqrt{2}}{3}$ C) ∞ D) 0
- 14) $\int_0^2 \frac{x}{\sqrt{4+x^2}} dx =$
- A) $2\sqrt{2}$ B) $2\sqrt{2} - 2$ C) $\sqrt{2} - 1$ D) $\frac{1}{2}(\sqrt{2} - 1)$
- 15) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be given by $f(x) = \sqrt{1 - 2x}$. The domain of f is
- A) $x \leq \frac{1}{2}$ B) $x \geq \frac{1}{2}$ C) $x \leq 2$ D) $x \geq 2$
- 16) The solution of the equation $|2x - 3| = 3$ is
- A) 0 B) 3 C) 0 and 3 D) -3
- 17) $\lim_{x \rightarrow 0} \frac{\sin px}{x} =$
- A) $\frac{1}{p}$ B) 1 C) $\frac{1}{2}$ D) P
- 18) Given that $y = \tanh 2x$, then $\frac{dy}{dx} =$
- A) $\text{sech}^2 2x$ B) $2\text{sech}^2 2x$ C) $\frac{1}{2}\text{sech}^2 2x$ D) $-2\text{sech}^2 2x$

19) The probability distribution of a discrete random variable X, the mean is

x	1	2	3
P(X=x)	1/2	1/3	1/6

- A) 1 B) 5/3 C) 1/3 D) 2/3

20) The probability density function of a continuous random variable X is

$$f(x) = \begin{cases} kx(2-x), & 0 \leq x \leq 2 \\ 0, & \text{elsewhere} \end{cases} . \text{ The value of K is}$$

- A) $\frac{3}{4}$ B) $\frac{4}{3}$ C) 0 D) 1

21) Two independent random variables X and Y are such that $E(X)=3$ and $E(Y)=4$, then $E(3X-2Y) =$

- A) 1 B) -1 C) 6 D) -8

22) Teams A has a probability $\frac{2}{3}$ of winning a game whenever it plays. If A plays 4 games, the probability that it wins exactly 2 games is :

- A) 1 B) $\frac{1}{3}$ C) $\frac{8}{27}$ D) $\frac{8}{9}$

23) Suppose 2% of the items made by a factory are defective. The probability P that there are 3 defective items in a sample of 100 items is

- A) 0.18 B) 0.02 C) 0.03 D) 0.135

24) A continuous random variable X has probability density function $f(x)$, where

$$f(x) = \begin{cases} k, & \text{for } 0 \leq x \leq 2 \\ k(2x-3), & \text{for } 2 \leq x \leq 3 \\ 0, & \text{elsewhere} \end{cases}$$

The value of k is

- A) $\frac{1}{4}$ B) $\frac{1}{2}$ C) 1 D) $\frac{3}{4}$

25) Given that the function $f: \mathbb{R} \rightarrow \mathbb{R}$ monotone increasing and that $\int_2^{15} f(x)dx = 27$

and $\int_2^9 f(x)dx = 19$, then $\int_9^{15} f(x)dx =$

- A) 46 B) 8 C) 6 D) 13

- 26) Let f and g be the function from the set of integers to itself, defined by $f(x) = 2x + 1$ and $g(x) = 3x + 4$. Then the composition of $(f \circ g)(x)$ is
 A) $6x + 9$ B) $6x + 7$ C) $6x + 6$ D) $6x + 8$
- 27) Given that $\ln x - \frac{6}{\ln x} + 1 = 0$, then the value of x are
 A) 2 and -3 B) 3 and -2 C) e^{-3} and e^2 D) e^3 and e^{-2}
- 28) If $x = \sec t$ and $y = \tan t$, then $\frac{dy}{dx}$ is
 A) $\sec t \tan t$ B) $\operatorname{cosec} t$ C) $\sec^2 t$ D) $\operatorname{cosec} t \cot t$
- 29) $\int \frac{2x}{(x^2+7)^6} dx =$
 A) $\frac{1}{6} \ln(x^2 + 7)$ B) $-\frac{1}{5(x^2 + 7)^5}$ C) $-\frac{1}{7(x^2 + 7)^7}$ D) $\frac{1}{3} \ln(x^2 + 7)$
- 30) $f(x, y) = \frac{2x+y}{1-xy}$, then the value of $f(1, -3)$ is
 A) -1 B) 0 C) $\frac{1}{4}$ D) $-\frac{1}{4}$
- 31) Let $f(x, y, z) = x^2 \ln yz + 2 \cos xy$. Then $f_x(x, y, z) =$
 A) $2x \ln yz - 2y \sin xy$ B) $2x \ln yz + 2y \sin xy$
 C) $2 \ln yz - 2x \sin xy$ D) $2 \ln yz + 2y \sin xy$
- 32) Let $f: \mathbb{R}^3 \rightarrow \mathbb{R}$ be given by $f(x, y, z) = x^2 + 3xy + yz^3$. Then $\operatorname{grad} f$ is
 A) $(5x, 3y, z^3 + 3yz^2)$ B) $(2x + 3y, 3x + z^3, 3yz^2)$
 C) $(x^2 + 3y, 3x + z^3, 3yz^2)$ D) $2x + 3y, 3x, z^3 + yz^3$
- 33) Given that $x^2y + 5x^3y^4 = 7$, $\frac{dy}{dx} =$
 A) $\frac{2xy+15x^2y^4}{x^2+20x^3y^3}$ B) $-\frac{2xy+15x^2y^4}{x^2+20x^3y^3}$ C) $\frac{-2xy+15x^2y^4}{x^2+20x^3y^3}$ D) $\frac{2xy+15x^2y^4}{-x^2+20x^3y^3}$
- 34) The value of x for which $\left(\frac{1}{25}\right)^x = 5^{2(x+3)}$ is
 A) $\frac{2}{3}$ B) $-\frac{2}{3}$ C) $\frac{3}{2}$ D) $-\frac{3}{2}$
- 35) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function defined by $f(x) = x^3 + 5x$. Then the function f is
 A) periodic B) even C) odd D) parity

36) The value of k for which the function $f: \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = \begin{cases} kx + 3, & \text{if } x < 2 \\ x^2 + 1, & \text{if } x \geq 2 \end{cases}$ is continuous at $x = 2$ is

- A) 5 B) 2 C) 1 D) -1

37) Let z be a complex number. The locus of points described by $|z + 2i| = 2$ is

- A) $x^2 + (y + 2)^2 = 2$ B) $x^2 + (y + 2)^2 = 4$
C) $y^2 + (x + 2)^2 = 2$ D) $y^2 + (x + 2)^2 = 4$

38) The principal argument of the complex number $z = \frac{-1 + \sqrt{3}i}{1 - i}$ is

- A) $\frac{11\pi}{12}$ B) $\frac{5\pi}{12}$ C) $\frac{7\pi}{12}$ D) $\frac{2\pi}{3}$

39) The value(s) of x for which $\log_3(x^2 - 10) - \log_3 x = 2$ is (are)

- A) 10 and -1 B) -10 and 1 C) -1 D) 10

40) If $y = \cosh 5x$ then $\frac{dy}{dx}$ is

- A) $-5 \sinh 5x$ B) $-\frac{1}{5} \sinh 5x$ C) $\frac{1}{5} \sinh 5x$ D) $5 \sinh 5x$

SECTION B: STRUCTURAL ANALYSIS : 30 MARKS

1) Let $f(x) = \frac{4\sqrt{x} - 4}{x^2 - 1}$.

- a) What is the domain of $f(x)$?
b) Find the limit

$$\lim_{x \rightarrow 1} \frac{4\sqrt{x} - 4}{x^2 - 1}$$

(5 marks)

2) Evaluate $\int \frac{\sqrt{x^2 - 9}}{x} dx$

(3 marks)

3) Solve the differential equation

a) $(y^2 - xy)dx + x^2 dy = 0.$

b) $\frac{d^2 y}{dt^2} + 3\frac{dy}{dt} + 2y = e^{-2t}$ given that $y=0$ and $dy/dt=0$ when $t=0$. (8 marks)

1) a) Find inverse Laplace transform, $L^{-1} \left\{ \frac{s+5}{(s^2-2s-3)} \right\}$ (4 marks)

b) Solve, using Laplace transformation, the differential equation

$y'' + 4y = e^{-t}$, $y(0) = 2$, $y'(0) = 1$ (5 marks)

2) Find the Fourier coefficients of the periodic function $f(x)$ given by $f(x) =$

$\begin{cases} -k & \text{if } -\pi < x < 0 \\ k & \text{if } 0 < x < \pi \end{cases}$ and $f(x + 2\pi) = f(x)$ (5 marks)

STATISTICS: 30 MARKS

1) The monthly earnings of computer engineers on part-time employment in a certain year in a firm are shown in the following table;

Earnings(millions)	1-3	4-6	7-9	10-12	13-15
Number of workers f	2	4	8	5	1

a) Calculate the mean, variance and standard deviation given your answers correct to 2 decimal places. (15 marks)

b) Calculate the median and mode of this data leaving your answers correct to 2 decimal places. (5 Marks)

2) In an experiment to determine the relationship between frequency and the inductive reactance of an electrical circuit, the following results were obtained:

Frequency(Hz)	Inductive reactance(ohm)
50	30
100	65
150	90
200	130
250	150
300	190
350	200

- 3) Determine the equation of the regression line of inductive reactance on frequency, assuming a linear relationship. **(5 marks)**
- 4) In an experiment to determine the relationship between frequency and the inductive reactance of an electrical circuit, the following results were obtained:

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350	200

Determine the equation of the regression line of inductive reactance on frequency, assuming a linear relationship. **(5 marks)**

PROBABILITY: 20 MARKS

1) Roll a single fair die and let X be the outcome. Suppose we want to compute the probability $p(x \geq 6)$ Calculate:

- a) The expected value $E(X)$
- b) The variance σ^2
- c) $p(x \geq 6)$ by exact probability
- d) $p(x \geq 6)$ by Markov inequality
- e) $p(x \geq 6)$ by the two-sided (usual) chebychev's inequality
- f) $p(x \geq 6)$ by the one-sided chebychev's inequality

(15 marks)

2) A shop stocks only two brands A and B of desktops. These desktops are of two sizes large (L) and small (S). 70% are of brand A and 30% of desktops are of brand B. 40% of those of brand B are of small size and 70% of brand A are of large size.

a) Draw a tree diagram to represent the above information. Find the probability that

b) A desktop chosen at random from the stock will be small size. **(2 marks)**

3) c) A small desktop chosen random from the stock will be of brand A. **(3 marks)**